

Lecture 6: More on Correlation vs. Causation

Last lecture we discussed omitted variable bias: a situation in which a variable that is not included as a regressor causes a failure of orthogonality. Today, we will discuss other common problems that can cause violations of the orthogonality condition. These failures of orthogonality cause OLS to be biased and inconsistent.

For simplicity, we will focus on the model with a single regressor,

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i.$$

However, each of the problems that we study will also cause bias and inconsistency in settings with many regressors.

Throughout, we will assume no perfect collinearity (in the single regressor case, this just means $\text{Var}(X_i) > 0$), i.i.d. sampling, and a large sample. This means that:

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(X_i, Y_i)}{\widehat{\text{Var}}(X_i)} \xrightarrow{p} \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)}$$

so the OLS estimator will converge to the population covariance over the population variance. From here, we can show that

$$\begin{aligned} \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)} &= \frac{\text{Cov}(X_i, \alpha + \beta \cdot X_i + \varepsilon_i)}{\text{Var}(X_i)} \\ &= \frac{\beta \cdot \text{Var}(X_i) + \text{Cov}(X_i, \varepsilon_i)}{\text{Var}(X_i)} \\ &= \beta + \frac{\text{Cov}(X_i, \varepsilon_i)}{\text{Var}(X_i)} \end{aligned}$$

This emphasizes that the issue of causation vs. correlation is not about the sample size: it is about whether orthogonality holds in the population. The key in each case will be to solve for $\text{Cov}(X_i, \varepsilon_i)$, which will show whether orthogonality holds. If $\text{Cov}(X_i, \varepsilon_i) \neq 0$, then we can also solve for the large-sample bias of OLS.

Measurement Error

Sometimes, we have only a noisy measure of the regressor. Suppose that the true regressor is X_i^* , but we observe $X_i = X_i^* + u_i$. We will focus on “classical” measurement error, which is the case where u_i is independent of X_i^* and ε_i^* , and $\mathbb{E}[u_i] = 0$.

The true causal model that we are interested in is

$$Y_i = \alpha + \beta \cdot X_i^* + \varepsilon_i^*$$

Suppose that ε_i^* has mean zero and is uncorrelated with X_i^* . This means that if we could observe X_i^* , then our estimated regression would be consistent. However, we instead run a regression with X_i . We can write the model with X_i to show what the error term, ε_i , is in that model.

$$\begin{aligned} Y_i &= \alpha + \beta \cdot X_i^* + \varepsilon_i^* \\ &= \alpha + \beta \cdot (X_i - u_i) + \varepsilon_i^* \\ &= \alpha + \beta \cdot X_i + \underbrace{(\varepsilon_i^* - \beta \cdot u_i)}_{\varepsilon_i} \end{aligned}$$

So, in the causal model with X_i , our error term is $\varepsilon_i = \varepsilon_i^* - \beta \cdot u_i$. Note that we have written this model so that α and β are the same coefficients that we are interested in from the model with X_i^* and ε_i^* . In order to do this, the measurement error, u_i , became part of the error term.

However, in this model, the orthogonality condition does not hold:

$$\begin{aligned}\text{Cov}(X_i, \varepsilon_i) &= \text{Cov}(X_i^* + u_i, \varepsilon_i^* - \beta \cdot u_i) \\ &= \text{Cov}(X_i^*, \varepsilon_i^* - \beta \cdot u_i) + \text{Cov}(u_i, \varepsilon_i^* - \beta \cdot u_i) \\ &= \text{Cov}(X_i^*, \varepsilon_i^*) - \beta \text{Cov}(X_i^*, u_i) + \text{Cov}(u_i, \varepsilon_i^*) - \beta \cdot \text{Cov}(u_i, u_i) \\ &= 0 - 0 + 0 - \beta \cdot \text{Var}(u_i) \neq 0\end{aligned}$$

Because the observed regressor, X_i , and the error term ε_i both contain the measurement error u_i , there is a correlation between the error term and the regressor.

We can also derive the variance of the regressor:

$$\begin{aligned}\text{Var}(X_i) &= \text{Var}(X_i^* + u_i) \\ &= \text{Cov}(X_i^* + u_i, X_i^* + u_i) \\ &= \text{Var}(X_i^*) + \text{Var}(u_i)\end{aligned}$$

We can then plug these formulas in to derive the large-sample bias in the OLS coefficient:

$$\begin{aligned}\hat{\beta} &\xrightarrow{p} \beta + \frac{\text{Cov}(X_i, \varepsilon_i)}{\text{Var}(X_i)} \\ &= \beta - \beta \cdot \frac{\text{Var}(u_i)}{\text{Var}(X_i^*) + \text{Var}(u_i)} \\ &= \beta \left(1 - \frac{\text{Var}(u_i)}{\text{Var}(X_i^*) + \text{Var}(u_i)} \right) \\ &= \beta \cdot \underbrace{\frac{\text{Var}(X_i^*)}{\text{Var}(X_i^*) + \text{Var}(u_i)}}_{\text{Signal Share: } \frac{\text{Signal}}{\text{Signal} + \text{Noise}}}\end{aligned}$$

Classical measurement error in the regressor thus leads to *attenuation*: the OLS estimate will be biased towards zero. The degree of attenuation depends on the ratio $\frac{\text{Var}(X_i^*)}{\text{Var}(X_i^*) + \text{Var}(u_i)}$. We can call this the “signal share,” connecting to the idea of signal vs. noise: $\text{Var}(X_i^*)$ reflects the signal in the regressor, while $\text{Var}(u_i)$ reflects the noise. If the variance of X_i is largely signal, then the attenuation bias will be low. But if much of the regressor is noise, then we will have a lot of attenuation.

What if we have measurement error in the outcome, Y_i ? Suppose now that we observe the true X_i , but we observe the outcome with noise: $Y_i = Y_i^* + u_i$, where Y_i^* is the true outcome variable. In the true causal model:

$$Y_i^* = \alpha + \beta \cdot X_i + \varepsilon_i^*$$

where ε_i^* is mean zero and uncorrelated with X_i . Plugging this into $Y_i = Y_i^* + u_i$, we then have:

$$Y_i = \alpha + \beta \cdot X_i + \underbrace{\varepsilon_i^* + u_i}_{\varepsilon_i}$$

Suppose again that the measurement error in the outcome is classical. This means that u_i is mean zero, and independent of X_i and of ε_i^* . We then have:

$$\begin{aligned}\text{Cov}(X_i, \varepsilon_i) &= \text{Cov}(X_i, \varepsilon_i^* + u_i) \\ &= \text{Cov}(X_i, \varepsilon_i^*) + \text{Cov}(X_i, u_i) \\ &= 0 + 0\end{aligned}$$

Classical measurement error in the outcome thus does not violate the orthogonality condition, and our OLS estimates will still be consistent in this scenario.

Note that classical measurement error is just one possibility: there are many types of non-classical measurement error, which can have different implications. In this course we will only cover classical measurement error, but if you are interested in learning more, you can refer to “Bias from Classical and Other Forms of Measurement Error” by Hyslop and Imbens (2001), and/or come ask me in office hours.

Reverse Causality and Simultaneity

Simultaneity occurs whenever the regressor and the outcome are jointly determined by a system of equations. Simultaneity can create feedback that causes the regressor to be correlated with the error term, violating orthogonality. We will consider two classic examples of simultaneity: demand estimation and reverse causality.

Supply and Demand

Suppose that we are trying to estimate the elasticity of the demand curve: how log price affects log quantity demanded. We can write this equation as

$$\log Q_i = \alpha_D + \beta_D \cdot \log P_i + \varepsilon_i^D$$

where $\log Q_i$ is log quantity, $\log P_i$ is log price, $\beta_D < 0$ is the price elasticity of demand, and ε_i^D is a demand shock. If we had exogenous variation in prices, then it would be straightforward to estimate this demand curve. However, prices are determined endogenously in equilibrium, based on both the demand curve and the supply curve. The supply curve is given by

$$\log Q_i = \alpha_S + \beta_S \cdot \log P_i + \varepsilon_i^S$$

where $\beta_S > 0$ is the price elasticity of supply and ε_i^S is a supply shock.

We will assume that ε_i^D and ε_i^S are uncorrelated and have mean zero:

$$\mathbb{E}[\varepsilon_i^D] = 0, \mathbb{E}[\varepsilon_i^S] = 0, \text{Cov}(\varepsilon_i^D, \varepsilon_i^S) = 0.$$

To study whether the regressor, $\log P_i$, is correlated with the error term in the demand equation, ε_i^D , it will be useful to solve for log price in terms of the supply and demand shocks. In equilibrium, quantity demanded equals quantity supplied, so we have:

$$\begin{aligned} \alpha_S + \beta_S \cdot \log P_i + \varepsilon_i^S &= \alpha_D + \beta_D \cdot \log P_i + \varepsilon_i^D \\ \implies (\beta_S - \beta_D) \cdot \log P_i &= \alpha_D - \alpha_S + \varepsilon_i^D - \varepsilon_i^S \\ \implies \log P_i &= \frac{\alpha_D - \alpha_S + \varepsilon_i^D - \varepsilon_i^S}{\beta_S - \beta_D} \end{aligned}$$

where the denominator $\beta_S - \beta_D$ is positive since the supply elasticity is positive and the demand elasticity is negative. Since price and quantity are determined simultaneously by the supply and demand equations, the log price is a function of both supply and demand shocks.

To study how this biases our OLS estimate of the demand elasticity, we can compute $\text{Cov}(\log P_i, \varepsilon_i^D)$ and $\text{Var}(\log P_i)$. We have:

$$\begin{aligned} \text{Cov}(\log P_i, \varepsilon_i^D) &= \text{Cov}\left(\frac{\alpha_D - \alpha_S + \varepsilon_i^D - \varepsilon_i^S}{\beta_S - \beta_D}, \varepsilon_i^D\right) \\ &= \frac{1}{\beta_S - \beta_D} \cdot \text{Cov}(\varepsilon_i^D - \varepsilon_i^S, \varepsilon_i^D) \\ &= \frac{1}{\beta_S - \beta_D} \cdot \text{Var}(\varepsilon_i^D) \\ \text{Var}(\log P_i) &= \text{Var}\left(\frac{\alpha_D - \alpha_S + \varepsilon_i^D - \varepsilon_i^S}{\beta_S - \beta_D}\right) \\ &= \frac{1}{(\beta_S - \beta_D)^2} \cdot \text{Var}(\varepsilon_i^D - \varepsilon_i^S) \\ &= \frac{1}{(\beta_S - \beta_D)^2} \cdot (\text{Var}(\varepsilon_i^D) + \text{Var}(\varepsilon_i^S)) \end{aligned}$$

We then have:

$$\begin{aligned}
 \hat{\beta}_D &\xrightarrow{p} \beta_D + \frac{\text{Cov}(\log P_i, \varepsilon_i^D)}{\text{Var}(\log P_i)} \\
 &= \beta_D + (\beta_S - \beta_D) \cdot \frac{\text{Var}(\varepsilon_i^D)}{\text{Var}(\varepsilon_i^D) + \text{Var}(\varepsilon_i^S)} \\
 &= \beta_D \cdot \underbrace{\frac{\text{Var}(\varepsilon_i^S)}{\text{Var}(\varepsilon_i^D) + \text{Var}(\varepsilon_i^S)}}_{\text{Share due to Supply Shocks}} + \beta_S \underbrace{\frac{\text{Var}(\varepsilon_i^D)}{\text{Var}(\varepsilon_i^D) + \text{Var}(\varepsilon_i^S)}}_{\text{Share due to Demand Shocks}}
 \end{aligned}$$

The formula shows that when variation in log price comes from both supply shocks and demand shocks, OLS will estimate a weighted average of the demand elasticity and the supply elasticity. The weights depend on what share of the variance in $\log P_i$ comes from supply shocks vs. demand shocks.

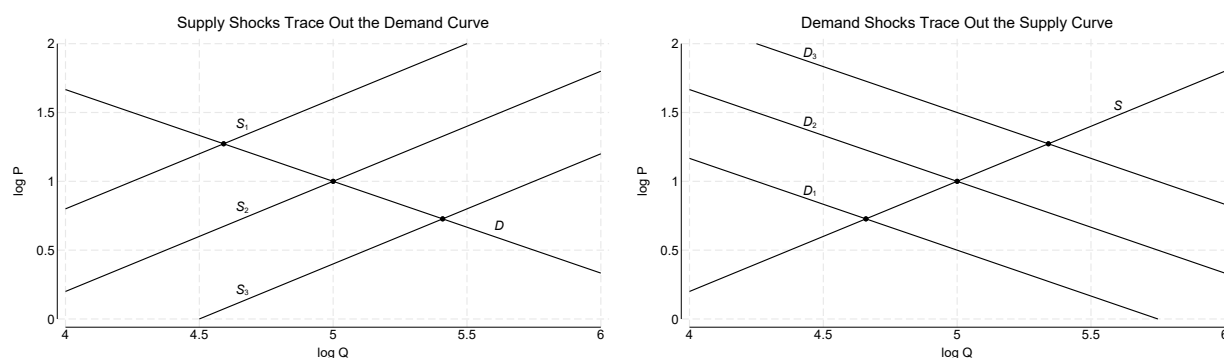


Figure 1: Estimation with Supply versus Demand Shocks

To understand this result better, Figure 1 shows equilibrium prices and quantities under different supply and demand shocks. In the left panel, the demand curve is held fixed and supply shocks shift the supply curve. In this case, shocks to supply create exogenous variation in price and trace out the demand curve. In the right panel, the supply curve is held fixed and demand shocks shift the demand curve. In this case, shocks to demand create variation in price that is correlated with the error term in the demand equation, and thus cannot be used to estimate the slope of the demand curve. Instead, demand shocks trace out the supply curve. Thus, unless all price variation comes from supply shocks, OLS estimation of the demand curve will generally be inconsistent.

Reverse Causality

We are generally interested in estimating the causal effect of X_i (the regressor) on Y_i (the outcome). But what happens if there is also a causal effect of Y on X ? Suppose that we have the following causal equations:

$$\begin{aligned}
 Y_i &= \alpha + \beta \cdot X_i + \varepsilon_i \\
 X_i &= \gamma + \delta \cdot Y_i + u_i
 \end{aligned}$$

where the top equation shows the causal effect of X on Y , while the bottom equation shows the causal effect of Y on X . We will suppose that ε_i and u_i are uncorrelated and both have mean zero. We will also assume that $|\delta \cdot \beta| < 1$. This ensures that the simultaneous equations have a unique solution, and that the solution is stable, meaning that the back-and-forth feedback between X_i and Y_i eventually dies out.

It will be useful to get an expression for X_i in terms of the error terms and coefficients. We have:

$$\begin{aligned} X_i &= \gamma + \delta \cdot Y_i + u_i \\ &= \gamma + \delta \cdot (\alpha + \beta \cdot X_i + \varepsilon_i) + u_i \\ \implies X_i - \delta \cdot \beta \cdot X_i &= \gamma + \delta \cdot \alpha + \delta \cdot \varepsilon_i + u_i \\ \implies X_i &= \kappa + \frac{\delta \cdot \varepsilon_i + u_i}{1 - \delta \cdot \beta} \end{aligned}$$

where $\kappa = \frac{\gamma + \delta \cdot \alpha}{1 - \delta \cdot \beta}$. We can once again test the orthogonality condition:

$$\begin{aligned} \text{Cov}(X_i, \varepsilon_i) &= \text{Cov}\left(\kappa + \frac{\delta \cdot \varepsilon_i + u_i}{1 - \delta \cdot \beta}, \varepsilon_i\right) \\ &= \frac{\delta}{1 - \delta \cdot \beta} \cdot \text{Var}(\varepsilon_i) \end{aligned}$$

Thus, there will be bias if $\delta \neq 0$. Since δ measures the causal effect of Y_i on X_i , this means that there will be bias whenever there is any reverse causality from the outcome to the regressor.

To compute the bias, we need to know the variance of X_i . We have:

$$\begin{aligned} \text{Var}(X_i) &= \text{Var}\left(\kappa + \frac{\delta \cdot \varepsilon_i + u_i}{1 - \delta \cdot \beta}\right) \\ &= \text{Var}\left(\frac{\delta \cdot \varepsilon_i}{1 - \delta \cdot \beta}\right) + \text{Var}\left(\frac{u_i}{1 - \delta \cdot \beta}\right) \\ &= \frac{\text{Var}(\delta \cdot \varepsilon_i) + \text{Var}(u_i)}{(1 - \delta \cdot \beta)^2} \end{aligned}$$

which we can plug in to get:

$$\begin{aligned} \hat{\beta} &\xrightarrow{p} \beta + \frac{\text{Cov}(X_i, \varepsilon_i)}{\text{Var}(X_i)} \\ &= \beta + (1 - \delta \cdot \beta) \frac{\delta \text{Var}(\varepsilon_i)}{\text{Var}(\delta \cdot \varepsilon_i) + \text{Var}(u_i)} \\ &= \beta \cdot \underbrace{\frac{\text{Var}(u_i)}{\text{Var}(\delta \cdot \varepsilon_i) + \text{Var}(u_i)}}_{\text{Share due to } u_i} + \frac{1}{\delta} \cdot \underbrace{\frac{\text{Var}(\delta \cdot \varepsilon_i)}{\text{Var}(\delta \cdot \varepsilon_i) + \text{Var}(u_i)}}_{\text{Share due to } \delta \cdot \varepsilon_i} \end{aligned}$$

where the last line is the result of some algebraic manipulations.

The formula shows that when there is reverse causality, our OLS estimate will be a weighted average of β , the causal effect of X_i on Y_i , and $\frac{1}{\delta}$, the inverse of the causal effect of Y_i on X_i .¹ The weights depend on what share of the variance of X_i comes from u_i versus $\delta \cdot \varepsilon_i$. Intuitively, if the variance of X_i comes largely from u_i , then OLS will mostly estimate the true causal effect β . But to the extent that the variance of X_i comes from $\delta \cdot \varepsilon_i$, this will cause OLS to mix in $\frac{1}{\delta}$, which comes from the reverse causality from Y_i to X_i .

To ensure a stable solution, we have assumed that $|\delta \cdot \beta| < 1$. Given this assumption, we can examine the bias term $(1 - \delta \cdot \beta) \frac{\delta \text{Var}(\varepsilon_i)}{\text{Var}(\delta \cdot \varepsilon_i) + \text{Var}(u_i)}$, and see that it will have the same sign as δ . Thus, reverse causality will cause OLS to be biased upward if δ is positive, and biased downward if δ is negative.

Sample Selection

We will finish with sample selection: the scenario in which we only observe a non-random subset of the population. When the sample is selected based on which observations survived (e.g. suppose we have performance data only for the hedge funds that are still open today), this is also called survivorship bias.

¹If we rearranged the second causal equation to put Y_i back on the left-hand side, the slope would be $\frac{1}{\delta}$.

We will suppose for simplicity that X_i and ε_i are independent. This implies orthogonality (independence implies uncorrelatedness), but is a stronger assumption. We will define the variable S_i , which is equal to one if observation i is included in the sample, and equal to zero otherwise. Since OLS is only run on the observations that are in the sample, our OLS estimate will equal

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(X_i, Y_i | S_i = 1)}{\widehat{\text{Var}}(X_i | S_i = 1)} \xrightarrow{p} \frac{\text{Cov}(X_i, Y_i | S_i = 1)}{\text{Var}(X_i | S_i = 1)} = \beta + \frac{\text{Cov}(X_i, \varepsilon_i | S_i = 1)}{\text{Var}(X_i | S_i = 1)}$$

When will $\text{Cov}(X_i, \varepsilon_i | S_i = 1)$ be equal to zero? One helpful fact is that:

$$\text{Cov}(X_i, \varepsilon_i | S_i = 1) = \text{Cov}(X_i, \mathbb{E}[\varepsilon_i | X_i, S_i = 1] | S_i = 1)$$

From here, we can ask: when will $\mathbb{E}[\varepsilon_i | X_i, S_i = 1]$ covary with X_i ? There are many types of selection, but one common form is selection on the outcome. Suppose that the outcome is *truncated from below*: we only observe it if Y_i is above a threshold. That is $S_i = 1$ if $Y_i > c$. We then have:

$$\begin{aligned} \mathbb{E}[\varepsilon_i | X_i, S_i = 1] &= \mathbb{E}[\varepsilon_i | X_i, Y_i > c] \\ &= \mathbb{E}[\varepsilon_i | X_i, \alpha + \beta \cdot X_i + \varepsilon_i > c] \\ &= \mathbb{E}[\varepsilon_i | \varepsilon_i > c - \alpha - \beta \cdot X_i] \end{aligned}$$

where the last line uses our assumption that X_i and ε_i are independent. This equation tells us that we will only observe the observation if its error term is above a cutoff value. As long as $\beta \neq 0$, that cutoff value will depend on X_i .

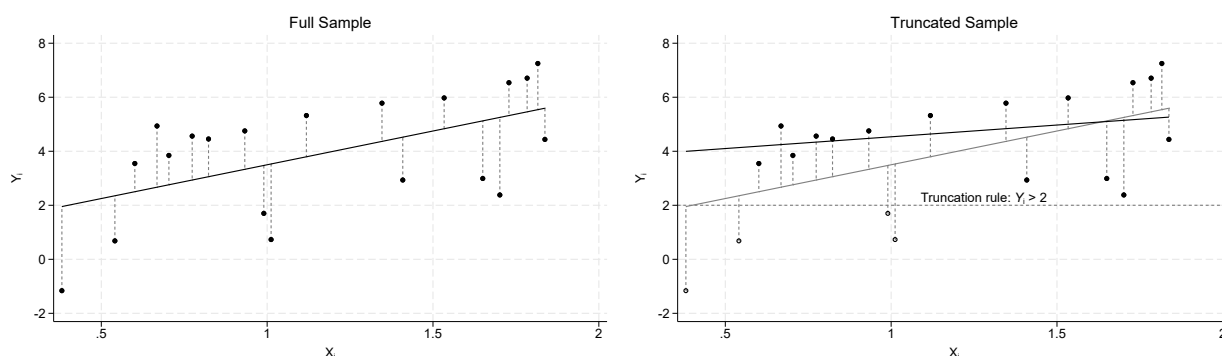


Figure 2: Regression Line Before and After Truncation of Y_i

To understand how this affects our estimates, Figure 2 plots a regression line before and after truncation from below. The left panel shows the full set of observations, the regression line before in solid black, and the error terms as deviations from that line. The right panel then truncates the data and drops all observations with $Y_i < 2$. The dropped observations are represented with hollow circles, the true regression line is now a solid gray line, and the new regression line is the solid black line. Notice that for low values of X_i , the dropped observations had negative draws of the error term ε_i , while the remaining observations had high draws of the error term. For high values of X_i , no observations are dropped. Since only high draws of the error term will survive truncation at low levels of X_i , truncation induces a correlation between X_i and ε_i that did not exist in the untruncated data. As a result, the new regression line is shallower than the original regression line.

In general, if β is positive then the cutoff will be decreasing in X_i , creating a negative covariance between X_i and ε_i in the selected sample. This will create a downward bias in OLS. If β is negative then the sign of the bias is reversed. Either way, truncation from below creates a bias towards zero. Similarly, we can show that truncation from above ($S_i = 1$ if $Y_i < c$) also creates bias towards zero. However, there are many different types of selection that can take place, which can bias OLS estimates in different directions: the key point to remember is that sample selection can create bias.